Project name:
Budget Forecasting & Cost Analysis using Predictive Analytics at OLIVE HOSPITALS

Submitted by:
Zainab Meher

Domain:
Data Analytics

Organisation:
Olive Hospital

Duration:
October 2025-May 2026

Project Overview

This project was carried out to analyze and forecast departmental expenditure patterns at Olive Sarvodaya Hospital, Hyderabad. The study focuses on understanding historical spending trends across key hospital departments and developing an interactive Power BI dashboard to support financial monitoring and decision-making.

The project combines data analysis, visualization, and forecasting techniques to identify expenditure patterns and estimate future costs. The findings provide insights into departmental spending behavior and can assist hospital management in budgeting and resource allocation.

OBJECTIVES OF THE STUDY

Objective 1: To Analyse Historical Expenditure Trends

To study and evaluate the monthly expenditure patterns of ICU, Radiology, and General Ward departments over the selected time period.

Objective 2: To Forecast Future Expenditure

To apply statistical forecasting techniques such as linear trend, moving average, and exponential smoothing to predict future departmental expenditure.

Objective 3: To Compare Departmental Performance

To compare expenditure levels and growth rates across departments in order to identify high-cost and high-growth areas.

Dataset Used

The dataset used in this project was obtained from departmental expenditure records maintained by Olive Sarvodaya Hospital.

Data Period:
January 2023 to December 2024

Departments Included:
• Intensive Care Unit (ICU)
• Radiology Department
• General Ward

Key Variables:
• Date
• Department Name
• Monthly Expenditure

The dataset contains monthly expenditure information that was used for trend analysis, comparison, and forecasting purposes.

Data Preparation and Transformation

Before visualization and forecasting, the dataset was reviewed and prepared for analysis.

The following data preparation activities were performed:

• Validation of departmental expenditure records.

• Standardization of date formats for time-series analysis.

• Verification of departmental classifications.

• Conversion of expenditure values into numerical format for calculations.

• Preparation of monthly expenditure data for dashboard visualization.

• Structuring data to support forecasting and trend analysis.

Tools and Technologies Used

Microsoft Excel
Used for preliminary data organization, validation, and forecasting calculations.

SQL
Used to extract, query, filter, and analyze hospital expenditure records from the database. SQL queries were utilized to retrieve department-wise expenditure data and support data analysis activities.

SQL queries were used to retrieve and organize departmental expenditure data from hospital records before further analysis and visualization.

Power BI
Used to develop interactive dashboards and visualizations for expenditure analysis and forecasting.

Power Query
Used for data cleaning, transformation, and preparation prior to dashboard development.

Statistical Techniques Applied

- Descriptive Trend Analysis
- Comparative Departmental Analysis
- Linear Trend Forecasting

Dashboard Components

The Power BI dashboard consists of the following components:

• Total Hospital Expenditure KPI

• ICU Expenditure KPI

• Radiology Expenditure KPI

• General Ward Expenditure KPI

• Monthly Expenditure Trend Analysis Chart

• Department-wise Expenditure Comparison Chart

• Department-wise Expenditure Distribution Pie Chart

• Forecast Visualization

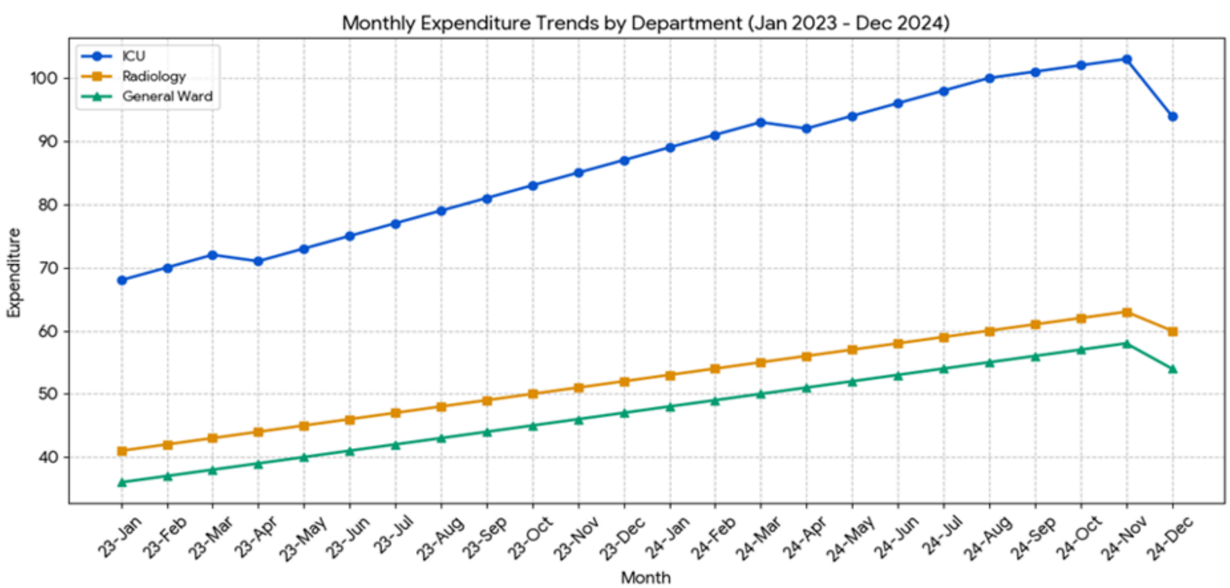
• Department Filter (Slicer)

These components were designed to provide both operational and strategic insights into hospital expenditure patterns.

Comparative Departmental Expenses

- ICU ranged from ₹68 to ₹103, giving an approximate central trend around ₹85–87 Lakhs
- Radiology ranged from ₹41 to ₹63, averaging around ₹51–53 Lakhs
- General Ward ranged from ₹36 to ₹58, averaging around ₹45–47 Lakhs

Considering the midpoint trend level as the representative historical



Comparative Forecast Summary (2025)

Department	Dec 2024	Dec 2025 Forecast	Approx Increase
ICU	94	122–123	~₹28 Lakhs
Radiology	60	~75	~₹15 Lakhs
General Ward	54	~69	~₹15 Lakhs

This comparison clearly indicates that ICU will impose the highest financial burden in the upcoming year.

Key Insights

• ICU recorded the highest expenditure among all departments and contributed approximately 46% of total hospital expenditure.

• Expenditure across all departments showed a consistent upward trend during the study period.

• Radiology emerged as the second-highest expenditure department with stable growth throughout the analysis period.

• General Ward displayed the lowest expenditure but maintained a steady increase over time.

• Forecast results indicate that departmental expenditure is expected to continue increasing in the future.

Forecasting Methodology

Linear trend forecasting techniques were applied to historical expenditure data to estimate future expenditure values.

The forecasting process involved:

• Analyzing historical monthly expenditure trends.

• Identifying growth patterns within each department.

• Applying trend-based forecasting calculations.

• Visualizing projected expenditure values through Power BI charts.

The forecasts provide an indication of future spending behavior and can support planning and budgeting activities.

Conclusion

The project successfully analyzed and visualized expenditure trends across major departments of Olive Sarvodaya Hospital. The findings indicate that ICU remains the largest contributor to hospital expenditure, while all departments demonstrate increasing spending trends over time. The developed dashboard provides a clear and interactive view of expenditure patterns and supports data-driven financial planning and decision-making within the healthcare environment.